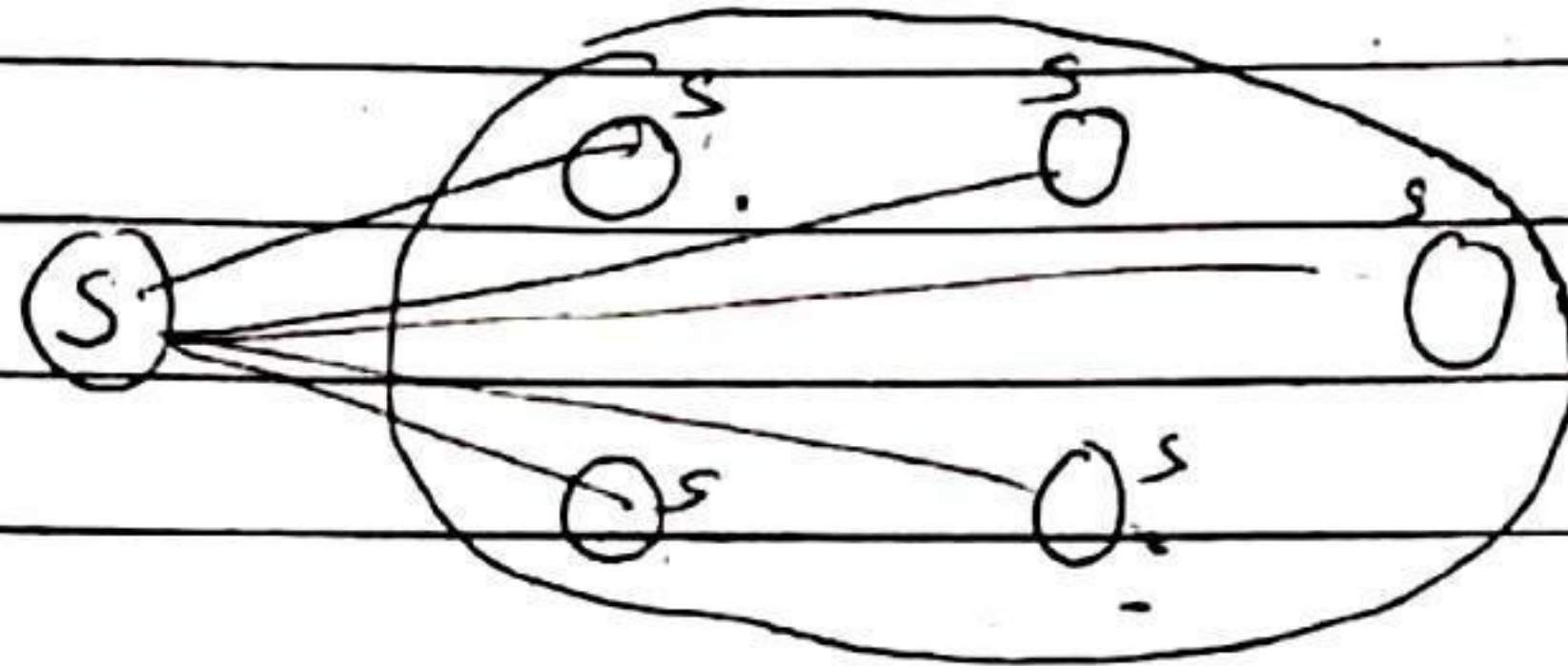


All pairs shortest path - Floyd Warshall.

- Relationship between single source shortest path and all pairs shortest path -



find shortest path b/w every two pairs.

Single source shortest path,

→ (S, a) - shortest path s to a
 (S, b) s to b
 (S, c) s to c
 $\vdots$

all pairs shortest path,

$(s, a) \rightarrow (a, b)$
 $(s, b) \rightarrow (a, c)$
 $(s, c) \rightarrow (b, c)$

→ you could run single source shortest path from every vertex then will become all pairs shortest path.

To implement SSSP (time)

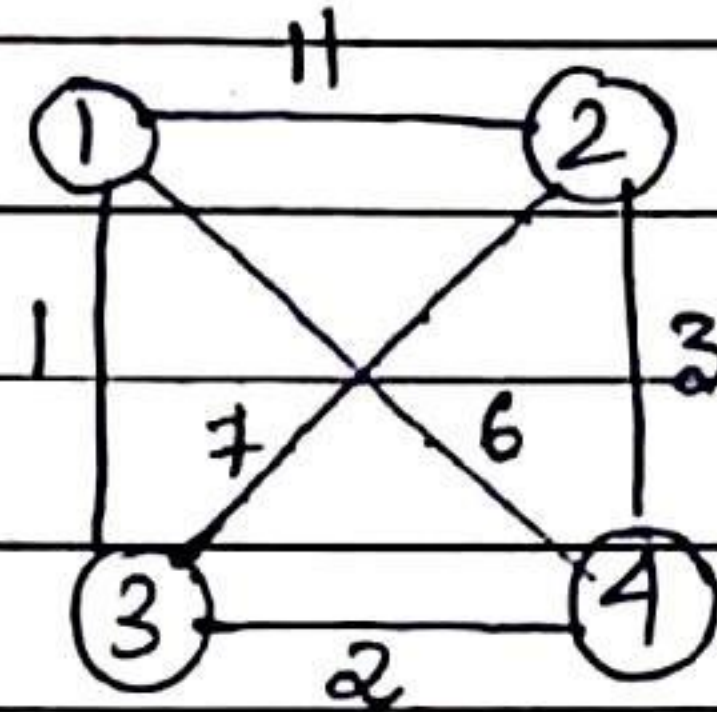
Dijkstra Algo - $O(E \log v)$

Bellman ford Algo - $O(vE)$

all pairs shortest path (time)

$O(v \cdot E \log v)$
 $= O(v \cdot v^2 \log v)$
 $= O(v^3 \log v)$

$O(v^2 E)$
 $= O(v^2 \cdot v^2) = O(v^4)$

Optimal SubstructureExample :

$$D =$$

	1	2	3	4
1	0	11	1	6
2	11	0	7	3
3	1	7	0	2
4	6	3	2	0

— path contain only one edge

$$D^1 =$$

	1	2	3	4
1	0	11	1	6
2	11	0	7	3
3	1	7	0	2
4	6	3	2	0

Distance of the path in which I might use 1 or might not use 1

$$(2,3) = 7 \checkmark$$

$$2 \rightarrow 1 \rightarrow 3 = 10$$

$$(2,4) = 3 \checkmark$$

$$2 \rightarrow 1 \rightarrow 4 = 17$$

$$D^2 =$$

	1	2	3	4
1	0	11	1	6
2	11	0	7	3
3	1	7	0	2
4	6	3	2	0

$$(1,3) = (1, 1 \rightarrow 2 + 2 \rightarrow 3) = 18$$

$$(1,4) = (6, 1 \rightarrow 2 + 2 \rightarrow 4) = 14$$

$$(4,3) = (2, 4 \rightarrow 2 + 2 \rightarrow 3) = 10$$

$$(2,3) = (7, 2 \rightarrow 1 + 1 \rightarrow 3) = 12$$

$$(3,4) = (2, 3 \rightarrow 2 + 2 \rightarrow 4) = 10$$

$D^3 =$ n^2

	1	2	3	4
1	0	8	1	3
2	8	0	7	3
3	1	7	0	2
4	3	3	2	0

vertex
can be include
 $\{1, 2, 3\}$

$$(1, 2) = (1, 1 \rightarrow 3 + 3 \rightarrow 2)$$

$8 \checkmark$

~~$$(1, 3) = (1, 1 \rightarrow 3 + 3 \rightarrow 1)$$~~

$$(1, 4) = (6, 1 \rightarrow 3 + 3 \rightarrow 4)$$

$1 + 2$
 $3 \checkmark$

~~$$(2, 3)$$~~

$$(2, 4) = (3, 2 \rightarrow 3 + 3 \rightarrow 4)$$

$7 \quad 2$
 9

~~$$(3, 4) = (4, 2)$$~~

$D^4 =$ n^2

	1	2	3	4
1	0	6	1	3
2	6	0	5	3
3	1	5	0	2
4	3	3	2	0

vertex
can be include
 $\{1, 2, 3, 4\}$

$$(1, 2) = (8, 1 \rightarrow 4 + 4 \rightarrow 2)$$

$3 + 3 = 6 \checkmark$

$$(1, 3) = (1, 1 \rightarrow 4 + 4 \rightarrow 3)$$

$3 + 2 = 5 \checkmark$

$$(2, 3) = (7, 2 \rightarrow 4 + 4 \rightarrow 3)$$

$3 + 2 = 5 \checkmark$

Subproblem on matrix * size of each one n^2

$$= n(n^2)$$

$$= O(n^3) \text{ — no. of subproblem.}$$

Total time complexity = $O(n^3)$
 space complexity = $O(n^2)$

Algorithm

~~FLO~~ FLOYD-WARSHALL (w) ^{weight matrix}

{

1) $n = w.rows$ // vertices.

2) $D^0 = w$

3) for $k=1$ to n // vertices

4) let $D^{(k)} = (d_{ij}^{(k)})$ be a ~~n x n~~ $n \times n$ matrix

5) for $i=1$ to n

6) for $j=1$ to n

7) $d_{ij}^{(k)} = \min \{ d_{ij}^{(k-1)}, d_{ik}^{(k-1)} + d_{kj}^{(k-1)} \}$

8) Return $D^{(n)}$

$$T(n) = O(n^3)$$

$$S(n) = O(n^3) = D^0 D^1 D^2 D^3 \dots D^n$$